

# Questions from 11th June 2025

## Question 1

On page 17, line 4-5, i write "From the geometry of  $K$  we see that if  $x, y \in V^*$  belongs to the same or adjacent  $(\mathcal{S}_{i_1} \circ \dots \circ \mathcal{S}_{i_n})(K)$  then we can connect  $x$  to  $y$  by one such chain". I still cannot figure out how we can connect through these chains.

Since by the restriction of  $x_{m+j} \underset{m+j+1}{\sim} x_{m+j+1}$  we cannot move very far in the fractal.

## Question 2

The whole of remark 2.28, there i am sure i misunderstood something. Especially with regards to  $P_t^m f(x)$ . But this also relates to the start of Section 2.4, where we say they are eigenfunctions, but end up using them as a basis.

## Question 3

In the proof of Theorem 2.25: How does the semigroup properties translate from  $Q_t$  to  $P_t$ . And also the same for Corollary 2.26

## Question 4

On page 43, why do we wish to show these FGF have an  $L^2$  density? And how does it follow from Proposition 3.11?

## Question 5

In the proof of Lemma 4.2 on page 47, we use <https://www.sciencedirect.com/science/article/pii/0022123669900317>[3, Theorem 2.1] but that requires  $\Delta_m$  to be infinitesimal gen of  $P_t^m$ , and the Theorem states there then exists a semigroup  $P_t$  on which  $P_t^m$  and  $P_t$  converge, but why do we know that it is the one from Theorem 2.25 of  $\Delta$ ?

## Smaller Questions:

1. On page 14, just under Definition 2.5, i have written that "In some cases we have that  $E_{m+n}(\tilde{u}) \leq r^n E_m(u)$  for some  $r \in (0, 1)$ ". But what cases are we talking about? This is also written on the bottom of page 10 in *Differential Equations on Fractals*.
2. Page 27, line 8, i write that  $U$  is a unitary map, and therefore a linear bijection. Is this correct? Link to source
3. Page 41, proof of lemma 3.6. Don't not get why the last equality holds since there will always be some real number left after multiplying the sums.
4. Page 43. Why is  $W(f) = \int_K f \, dW$  a well defined gaussian random variable. And why does it give rise to a isonormal. And why do we want this.

## Mistakes

1. Page 4, line 2. "... for each  $x \in F$ ..." should be "... for each  $x \in K$ ..."
2. Page 6, Lemma 1.10, line 1. "... such that  $\mu(F) \in (0, \infty)$ ..." should be "... such that  $\mu(K) \in (0, \infty)$ ..."
3. Page 6, Lemma 1.11, line 3. "... intersects at most  $(1 + 2a_2)^n a_1^n$ ..." should be "... intersects at most  $(1 + 2a_2)^n a_1^{-n}$ ..."
4. Page 8, Line 9 from the bottom. " $x_{i_1, \dots, i_k} \in K \cap B$  ..." should be " $x_{i_1, i_2 \dots} \in K \cap B$  ...".
5. Page 14, Definition 2.5, line 1. "Let  $\tilde{u} : V_{m+n} \rightarrow \mathbb{R}$  with  $\tilde{u}|_{V_m} = u$  being a *harmonic function* if ..." should be "Let  $\tilde{u} : V_{m+n} \rightarrow \mathbb{R}$  with  $\tilde{u}|_{V_m} = u$ . We call  $\tilde{u}$  a *harmonic function* if ...".
6. Page 16, Definition 2.8, line 4. " $\text{dom } \mathcal{E} = \{u \in \ell(V_*) : \mathcal{E}(u) < \infty\}$ " should be " $\text{dom } \mathcal{E} = \{u \in \ell(V^*) : \mathcal{E}(u) < \infty\}$ ".
7. Page 17, line 16. "The constants  $3^m$  and  $5^m$  are not random, ..." should be "The constants  $3^m$  and  $(N_p)^{-1}$  from  $\Delta_m$  are not random, ...".
8. Page 17, line 4 from the bottom. "... the values in  $K \setminus V_m$  comes from ..." should be "... the values in  $V^* \setminus V_m$  comes from ...".
9. Page 18, line 11 from the bottom. "... Then  $F_m = K \setminus \left(\bigcup_{\alpha \in \Omega_m} \overline{B}_\alpha\right)$ ." should be "... Then  $F_m = K \setminus \left(\bigcup_{\alpha \in \Omega_m} \overline{T}_\alpha\right)$ ..".
10. Page 25, Definition 2.21, line 2. "... the *discrete fractional Laplacian*  $(-\Delta_m)^s : L^2(V_m, \mu_m) \rightarrow L^2(V_m, \mu_m)$  ..." should be "... the *discrete fractional Laplacian*  $(-\Delta_m)^{-s} : L^2(V_m, \mu_m) \rightarrow L^2(V_m, \mu_m)$  ...".
11. Page 29, line 13. "... generator of a strongly continuous Markov semigroup" should be "... generator of a strongly continuous self-adjoint contraction semigroup".
12. Page 33, line 1 from the bottom. " $|\phi_j| \leq C e^{\lambda_j/4}$ " should be " $|\phi_j(x)| \leq C e^{\lambda_j/4}$ ".
13. Page 37, line 1. "Now choose  $k'$  such  $u_{k'} \leq R$  ..." should be "Now choose  $k'$  such  $u_{k'} \geq R$  ...".
14. Page 37, line 5. "We can likewise bound  $G_s(x, y)$  by" should be "We can likewise bound  $G_s(x, y)$  for  $s < \frac{d_h}{d_w}$  by".
15. Page 41, line 5 from the bottom. "and  $\phi$  is positive semidefinite" should be "and  $\Phi$  is positive semidefinite"
16. Page 50, line 6. "... the characteristic functional, e.i. that ..." should be "... the characteristic functional, i.e. that ..."
17. Page 58, line 4. "... what values the simulation converges to." should be "... what values the simulation diverges to."